

IIT JAM 2010 Official Paper

Category: Classic Era (2005–2014) • Official Mock Test Series

TOTAL QUESTIONS

15

TOTAL MARKS

90 M

TEST DURATION

60 Min

PAPER PATTERN

MCQ

Section / Type		Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)		15	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2010 • Q1 • ID: JAM_2010_Q1)

MCQ

+6 M

Neg: -2.0

Q.1 Which of the following conditions does NOT ensure the convergence of a real sequence $\{a_n\}$?

- (A) $|a_n - a_{n+1}| \rightarrow 0$ as $n \rightarrow \infty$
- (B) $\sum_{n=1}^{\infty} |a_n - a_{n+1}|$ is convergent
- (C) $\sum_{n=1}^{\infty} n a_n$ is convergent
- (D) The sequences $\{a_{2n}\}$, $\{a_{2n+1}\}$ and $\{a_{3n}\}$ are convergent

Question 2 (JAM 2010 • Q2 • ID: JAM_2010_Q2)

MCQ

+6 M

Neg: -2.0

Q.2 The value of $\iint_G \frac{\ln(x^2 + y^2)}{x^2 + y^2} dx dy$, where $G = \{(x, y) \in \mathbf{R}^2 : 1 \leq x^2 + y^2 \leq e^2\}$, is

- (A) π (B) 2π (C) 3π (D) 4π

Question 3 (JAM 2010 • Q3 • ID: JAM_2010_Q3)

MCQ

+6 M

Neg: -2.0

Q.3 The number of elements of S_5 (the symmetric group on 5 letters) which are their own inverses equals

- (A) 10 (B) 11 (C) 25 (D) 26

Question 4 (JAM 2010 • Q4 • ID: JAM_2010_Q4)

MCQ

+6 M

Neg: -2.0

Q.4 Let S be an infinite subset of $\mathbf{R}$ such that $S \cap \mathbf{Q} = \emptyset$. Which of the following statements is true?

- (A) S must have a limit point which belongs to $\mathbf{Q}$
- (B) S must have a limit point which belongs to $\mathbf{R} \setminus \mathbf{Q}$
- (C) S cannot be a closed set in $\mathbf{R}$
- (D) $\mathbf{R} \setminus S$ must have a limit point which belongs to S

Question 5 (JAM 2010 • Q5 • ID: JAM_2010_Q5)

MCQ

+6 M

Neg: -2.0

Q.5 Let $f: (1, 4) \rightarrow \mathbf{R}$ be a uniformly continuous function and let $\{a_n\}$ be a Cauchy sequence in $(1, 2)$. Let $x_n = a_n^2 f(a_n^2)$ and $y_n = \frac{1}{1+a_n^2} f(a_n^2)$, for all $n \in \mathbf{N}$. Which of the following statements is true?

- (A) Both $\{x_n\}$ and $\{y_n\}$ must be Cauchy sequences in $\mathbf{R}$
- (B) $\{x_n\}$ must be a Cauchy sequence in $\mathbf{R}$ but $\{y_n\}$ need not be a Cauchy sequence in $\mathbf{R}$
- (C) $\{y_n\}$ must be a Cauchy sequence in $\mathbf{R}$ but $\{x_n\}$ need not be a Cauchy sequence in $\mathbf{R}$
- (D) Neither $\{x_n\}$ nor $\{y_n\}$ needs to be a Cauchy sequence in $\mathbf{R}$

Question 6 (JAM 2010 • Q6 • ID: JAM_2010_Q6)

MCQ

+6 M

Neg: -2.0

Q.6 Let $\vec{F} = 2xyz e^{x^2} \hat{i} + z e^{x^2} \hat{j} + y e^{x^2} \hat{k}$ be the gradient of a scalar function. The value of $\int_L \vec{F} \cdot d\vec{r}$ along the oriented path L from $(0, 0, 0)$ to $(1, 0, 2)$ and then to $(1, 1, 2)$ is

- (A) 0
- (B) $2e$
- (C) e
- (D) e^2

Question 7 (JAM 2010 • Q7 • ID: JAM_2010_Q7)

MCQ

+6 M

Neg: -2.0

Q.7 Let $\vec{F} = xy \hat{i} + y \hat{j} - yz \hat{k}$ denote the force field on a particle traversing the path L from $(0, 0, 0)$ to $(1, 1, 1)$ along the curve of intersection of the cylinder $y = x^2$ and the plane $z = x$. The work done by $\vec{F}$ is

- (A) 0
- (B) $\frac{1}{4}$
- (C) $\frac{1}{2}$
- (D) 1

Question 8 (JAM 2010 • Q8 • ID: JAM_2010_Q8)

MCQ

+6 M

Neg: -2.0

Q.8 Let $\mathbf{R}[X]$ be the ring of real polynomials in the variable X . The number of ideals in the quotient ring $\mathbf{R}[X]/(X^2 - 3X + 2)$ is

- (A) 2
- (B) 3
- (C) 4
- (D) 6

Question 9 (JAM 2010 • Q9 • ID: JAM_2010_Q9)

MCQ

+6 M

Neg: -2.0

Q.9 Consider the differential equation $\frac{dy}{dx} = ay - by^2$, where $a, b > 0$ and $y(0) = y_0$. As $x \rightarrow +\infty$, the solution $y(x)$ tends to

(A) 0

(B) $\frac{a}{b}$ (C) $\frac{b}{a}$ (D) y_0

Question 10 (JAM 2010 • Q10 • ID: JAM_2010_Q10)

MCQ

+6 M

Neg: -2.0

Q.10 Consider the differential equation $(x+y+1)dx + (2x+2y+1)dy = 0$. Which of the following statements is true?

(A) The differential equation is linear

(B) The differential equation is exact

(C) e^{x+y} is an integrating factor of the differential equation

(D) A suitable substitution transforms the differentiable equation to the variables separable form

Question 11 (JAM 2010 • Q11 • ID: JAM_2010_Q11)

MCQ

+6 M

Neg: -2.0

Q.11 Let $T: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be a linear transformation such that $T((1, 2)) = (2, 3)$ and $T((0, 1)) = (1, 4)$. Then $T((5, 6))$ is

(A) $(6, -1)$ (B) $(-6, 1)$ (C) $(-1, 6)$ (D) $(1, -6)$

Question 12 (JAM 2010 • Q12 • ID: JAM_2010_Q12)

MCQ

+6 M

Neg: -2.0

Q.12 The number of 2×2 matrices over $\mathbf{Z}_3$ (the field with three elements) with determinant 1 is

(A) 24

(B) 60

(C) 20

(D) 30

Question 13 (JAM 2010 • Q13 • ID: JAM_2010_Q13)

MCQ

+6 M

Neg: -2.0

Q.13 The radius of convergence of the power series $\sum_{n=0}^{\infty} a_n z^{n^2}$, where $a_0 = 1$, $a_n = 3^{-n} a_{n-1}$ for $n \in \mathbf{N}$, is

- (A) 0 (B) $\sqrt{3}$ (C) 3 (D) ∞

Question 14 (JAM 2010 • Q14 • ID: JAM_2010_Q14)

MCQ

+6 M

Neg: -2.0

Q.14 Let $T : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be the linear transformation whose matrix with respect to the standard basis $\{e_1, e_2, e_3\}$ of $\mathbf{R}^3$ is $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$. Then T

- (A) maps the subspace spanned by e_1 and e_2 into itself
 (B) has distinct eigenvalues
 (C) has eigenvectors that span $\mathbf{R}^3$
 (D) has a non-zero null space

Question 15 (JAM 2010 • Q15 • ID: JAM_2010_Q15)

MCQ

+6 M

Neg: -2.0

Q.15 Let $T : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be the linear transformation whose matrix with respect to the standard basis of $\mathbf{R}^3$ is $\begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix}$, where a, b, c are real numbers not all zero. Then T

- (A) is one-to-one
 (B) is onto
 (C) does not map any line through the origin onto itself
 (D) has rank 1

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2010 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+6	A
2	MCQ	+6	B
3	MCQ	+6	D
4	MCQ	+6	D
5	MCQ	+6	A
6	MCQ	+6	B
7	MCQ	+6	C
8	MCQ	+6	C

Q. No.	Type	Marks	Official Key
9	MCQ	+6	B
10	MCQ	+6	D
11	MCQ	+6	A
12	MCQ	+6	A
13	MCQ	+6	B
14	MCQ	+6	C
15	MCQ	+6	C

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.